

# SAYAN KARMAKAR

## PRODUCT DEVELOPMENT ENGINEER

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### SUMMARY

Python backend engineer with experience developing enterprise-grade HR workflows and data-driven features for internal platforms at a global organization. Experienced in building RESTful APIs using FastAPI and Django, designing relational data models, and implementing authentication and role-based access. Strong backend fundamentals with a focus on clean, modular, and scalable systems.

### TECHNICAL SKILLS

**Languages:** Python, SQL, Java  
**Databases:** PostgreSQL, MySQL  
**Cloud:** AWS (EC2, S3, Route 53)

**Frameworks:** FastAPI, Django, DRF  
**Authentication:** JWT, OAuth  
**Tools:** Git, Docker, Jenkins, Swagger, Postman

### PROFESSIONAL EXPERIENCE

#### Product Development Engineer, ADP

*Aug 2022 – Present*

- Developed enterprise HR workflows and data pipelines for an internal ADP platform, implementing complex business rules, validations, and role-based access across multiple modules.
- Designed and managed relational data models for HR-related entities, ensuring data integrity, efficient querying, and seamless integration across internal services.
- Worked with containerized development environments and CI workflows to maintain consistency across development, testing, and deployment stages.
- Collaborated closely with product managers, QA engineers, and business stakeholders to translate functional requirements into scalable backend features.
- Actively contributed to debugging, enhancement, and optimization of existing modules to improve reliability and performance of internal systems.

### PROJECTS

#### Recipe Management REST API — [link](#)

- Built a production-ready backend API using Django REST Framework for recipe and ingredient management.
- Implemented JWT authentication, role-based access, and interactive API documentation via Swagger.
- Integrated image upload and storage functionality, enabling users to attach media to recipes and improving engagement.
- **Tech Stack:** Django REST Framework, PostgreSQL, JWT, Swagger.

#### Tunify – Music Streaming Backend Application (In Progress) — [link](#)

- Developed a scalable music streaming backend using FastAPI, PostgreSQL, and Async SQLAlchemy with JWT-based authentication and role-based access control.
- Implemented core features including user management, artists, albums, tracks, and playlist creation using clean service-repository architecture.
- Designed a modular, production-ready system with API versioning, Alembic migrations, centralized error handling, and Dockerized deployment.
- **Tech Stack:** FastAPI, Python, PostgreSQL, SQLAlchemy (Async), Alembic, JWT, Docker.

### EDUCATION

**B.Tech in Computer Science** - Assam University

*2022*

CGPA: 7.48 / 10

**Class XII** - West Bengal Board

*2017*

77.6%

**Class X** - West Bengal Board

*2015*

77.5%

### ADDITIONAL INFORMATION

**Languages:** English, Hindi, Bengali

**Interests:** Outdoor sports, Music